Problem 1: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
   int i= 1;

   int s =1;

   while(s <= n)
   {
      i++;
      s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer:(penalty regime: 0 %)

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Answer:

```
#include <stdio.h>


void function(int n)
```

```
{
   int counter = 0;

   int i = 1;
   counter++;        // i = 1

   int s = 1;
   counter++;        // s = 1

   while (s<=n)
   {
      counter++;     // while(s <= n)

      i++;
      counter++;     // i++

      s += i;
      counter++;     // s += i
   }
   counter++;

   printf("%d", counter);
}

int main()
{
   int n;
   scanf("%d", &n);
```

```
    function(n);


    return 0;
}
```

OP:

	Input	Expected	Got	
	9	12	12	
	4	9	9	

Passed all tests!

Correct